

# MOS IMP

## Ch-2

Stress (stress):

$$\sigma = \frac{P}{A} = \frac{\text{Force}}{\text{Area}}$$

Strain (strain):

$$\epsilon = \frac{\Delta l}{l} = \frac{\text{Change in length}}{\text{Original length}}$$

Change in length:

$$E = \frac{\sigma}{\epsilon}$$

$$\Delta l = \frac{P \cdot l}{A E}$$

$$\therefore \Delta l = E \cdot \epsilon$$

Hook's law:

$$A = \frac{\pi}{4} \times d^2$$

$$E = \frac{\sigma}{\epsilon}$$

$$E = 2 \times 10^5$$

(SUB-2)

Ex-2  $d = 12 \text{ mm}$

$l = 1.5 \text{ m} = 1500 \text{ mm}$

$P = 30 \text{ kN} = 30 \times 10^3 \text{ N}$

$E = 2 \times 10^5 \text{ N/mm}^2$

Find:  $\sigma$

$E$

change in length:

$\Delta l$

Ans:  $\sigma = \frac{P}{A}$

$$A = \frac{\pi}{4} \times d^2$$

$$= \frac{30 \times 10^3}{113.09} = 265.27$$

$$= \frac{\pi}{4} \times 12^2 = 113.09 \text{ mm}^2$$

$$\sigma = 265.27 \text{ N/mm}^2$$

$$\epsilon = \frac{\sigma}{E} = \frac{265.27}{(2 \times 10^5)}$$

$$\epsilon = 1.326 \times 10^{-3}$$

change in length  $\Delta l = E \cdot \epsilon$

$$= 1.326 \times 10^{-3} \times 1500$$

$$\Delta l = 1.989 \text{ mm}$$

Final length =  $1500 + 1.989$

$$= 1501.989 \text{ mm}$$

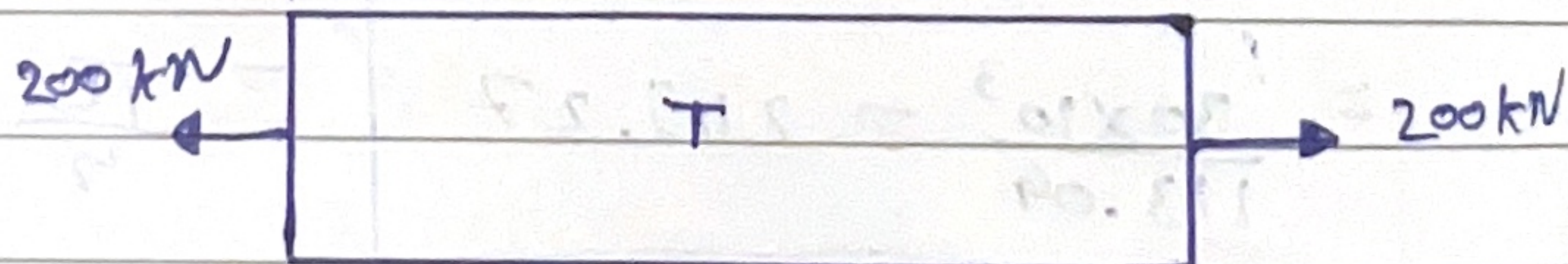
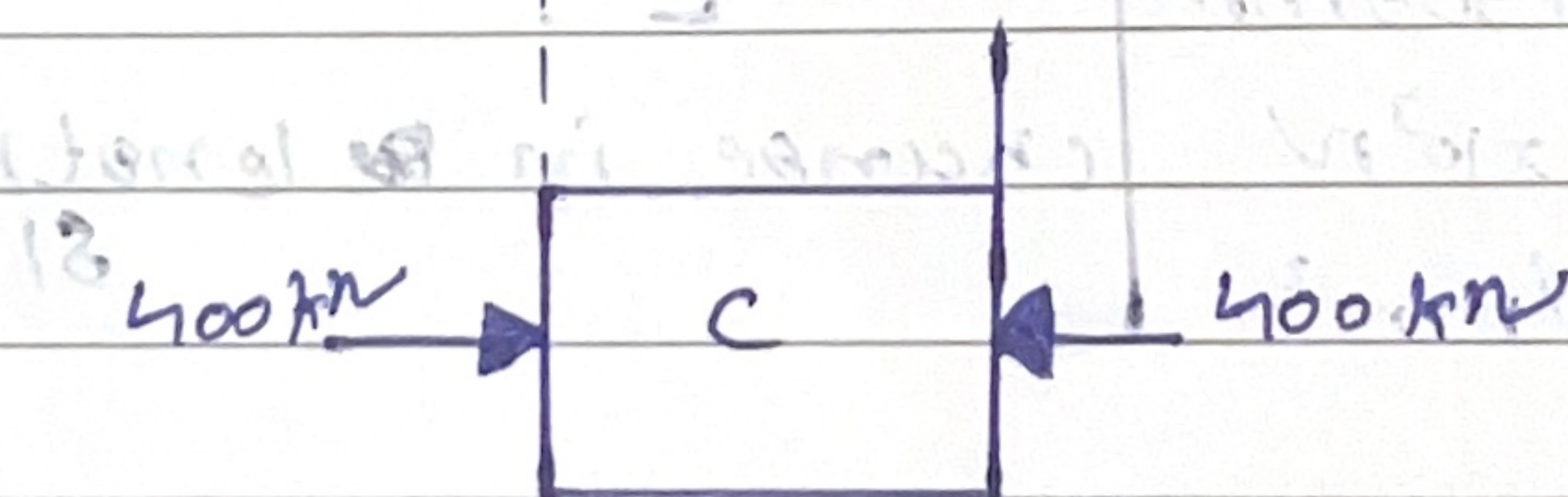
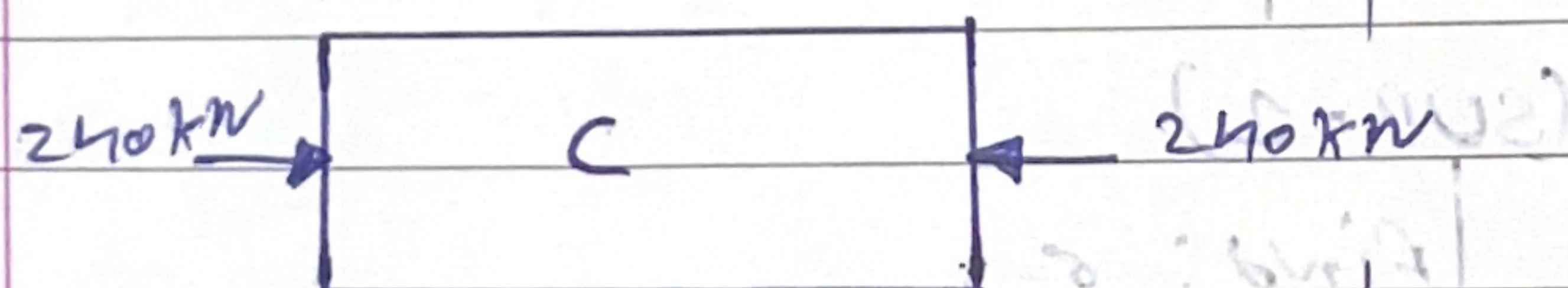
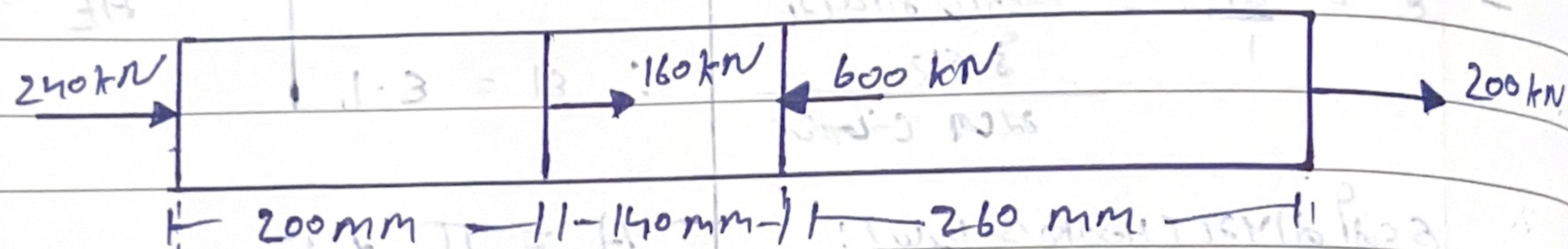


## SUBTOPIC - 2

For SUB-2  $\Delta l = \pm \delta l_1 \pm \delta l_2 \pm \delta l_3$

$$\Delta l = \pm \frac{P_1 l_1}{A_1 E_1} \pm \frac{P_2 l_2}{A_2 E_2} \pm \frac{P_3 l_3}{A_3 E_3}$$

Ex-1



1. Given Data:

$$E = 2 \times 10^5$$

$$A = 400 \text{ mm}^2$$



$$P_1 = -240 \text{ kN}$$

$$l_1 = 200 \text{ mm}$$

$$P_2 = -400 \text{ kN}$$

$$l_2 = 140 \text{ mm}$$

$$P_3 = 200 \text{ kN}$$

$$l_3 = 260 \text{ mm}$$

$$\therefore \delta l = \frac{P_1 l_1}{AE} \pm \frac{P_2 l_2}{AE} \pm \frac{P_3 l_3}{AE}$$

$$= \frac{1}{AE} \times [P_1 l_1 \pm P_2 l_2 \pm P_3 l_3]$$

$$kN =$$

$$= \frac{1}{400 \times 2 \times 10^5} \times [-240 \times 200 - 400 \times 140 + 200 \times 260]$$

$$N =$$

$$= \frac{1}{400 \times 2 \times 10^5} \times [-240 \times 10^3 \times 200 - 400 \times 10^3 \times 140 + 200 \times 10^3 \times 260]$$

$$= \frac{1}{400 \times 2 \times 10^5} \times [-52 \times 10^6] = \frac{(-52 \times 10^6)}{(80 \times 10^6)}$$

$$\delta l = -0.65$$

$$S_{min} P.F. = 0.855 = 0.855$$

$$0.855 = \frac{S}{S_y} = \frac{S}{200}$$

$$P.F. = 0.855 = \frac{S}{200} \times \frac{1}{1.2}$$

$$S_{min} P.F. = 0.855 = 0.855$$



SUB-3

- For SUB-3 MIMP:

$$\frac{\sigma_s}{E_s} = \frac{\sigma_c}{E_c} \quad \text{--- ①}$$

$$P = P_s + P_c$$

$$\therefore P = \sigma_s \cdot A_s + \sigma_c \cdot A_c \quad \text{--- ②}$$

Ex-2 Given Data:

- dia. of column = 250 mm
- ~~dia of~~ 6 bars of 22 mm diameter
- $P = 875 \text{ kN}$
- $E_s = 2.1 \times 10^5 \text{ N/mm}^2$ ,  $E_c = 0.14 \times 10^5 \text{ N/mm}^2$

Find: stress in steel ( $\sigma_s$ )stress in concrete ( $\sigma_c$ )

$$\text{Ans) } A_s = 6 \times \frac{\pi}{4} \times d_s^2$$

$$= 6 \times \frac{\pi}{4} \times (22)^2$$

$$A_s = 2280.79 \text{ mm}^2$$

$$A_c = \frac{\pi}{4} \times d_c^2 - A_s$$

$$= \frac{\pi}{4} \times 250^2 - 2280.79$$

$$A_c = 46806.59 \text{ mm}^2$$



we know that :

$$\frac{\sigma_s}{E_s} = \frac{\sigma_c}{E_c}$$

$$\therefore \sigma_s = \frac{E_s}{E_c} \times \sigma_c$$

$$= \frac{(2.1 \times 10^5)}{(0.14 \times 10^5)} \times \sigma_c$$

$$\sigma_s = 15 \sigma_c$$

$$P = P_s + P_c$$

$$P = \sigma_s A_s + \sigma_c A_c$$

$$875 \times 10^3 = 15 \sigma_c \times 2280.79 + \sigma_c \times 46806.59$$

$$875 \times 10^3 = 81018.44 \sigma_c$$

$$\sigma_c = \frac{875 \times 10^3}{81018.44}$$

$$\sigma_c = 10.80 \text{ N/mm}^2 \text{ stress in concrete}$$

$$\therefore \sigma_s = 15 \sigma_c$$

$$= 15 \times 10.80$$

$$\sigma_s = 162 \text{ N/mm}^2 \text{ stress in steel}$$



Ch-2

IMP for Moment of Inertia:

$$\bar{x} = \frac{a_1 \cdot x_1 + a_2 \cdot x_2 + a_3 \cdot x_3}{a_1 + a_2 + a_3}$$

$$\bar{y} = \frac{a_1 \cdot y_1 + a_2 \cdot y_2 + a_3 \cdot y_3}{a_1 + a_2 + a_3}$$

$$I_{xx1} = I_0 + ah^2$$

$$= \frac{BD^3}{12} + A x (y_1 - \bar{y})^2$$

$$\therefore h = y_1 - \bar{y}$$

$$I_{xx2} = I_0 + ah^2$$

$$= \frac{DB^3}{12} + A x (\bar{y} - y_2)^2$$

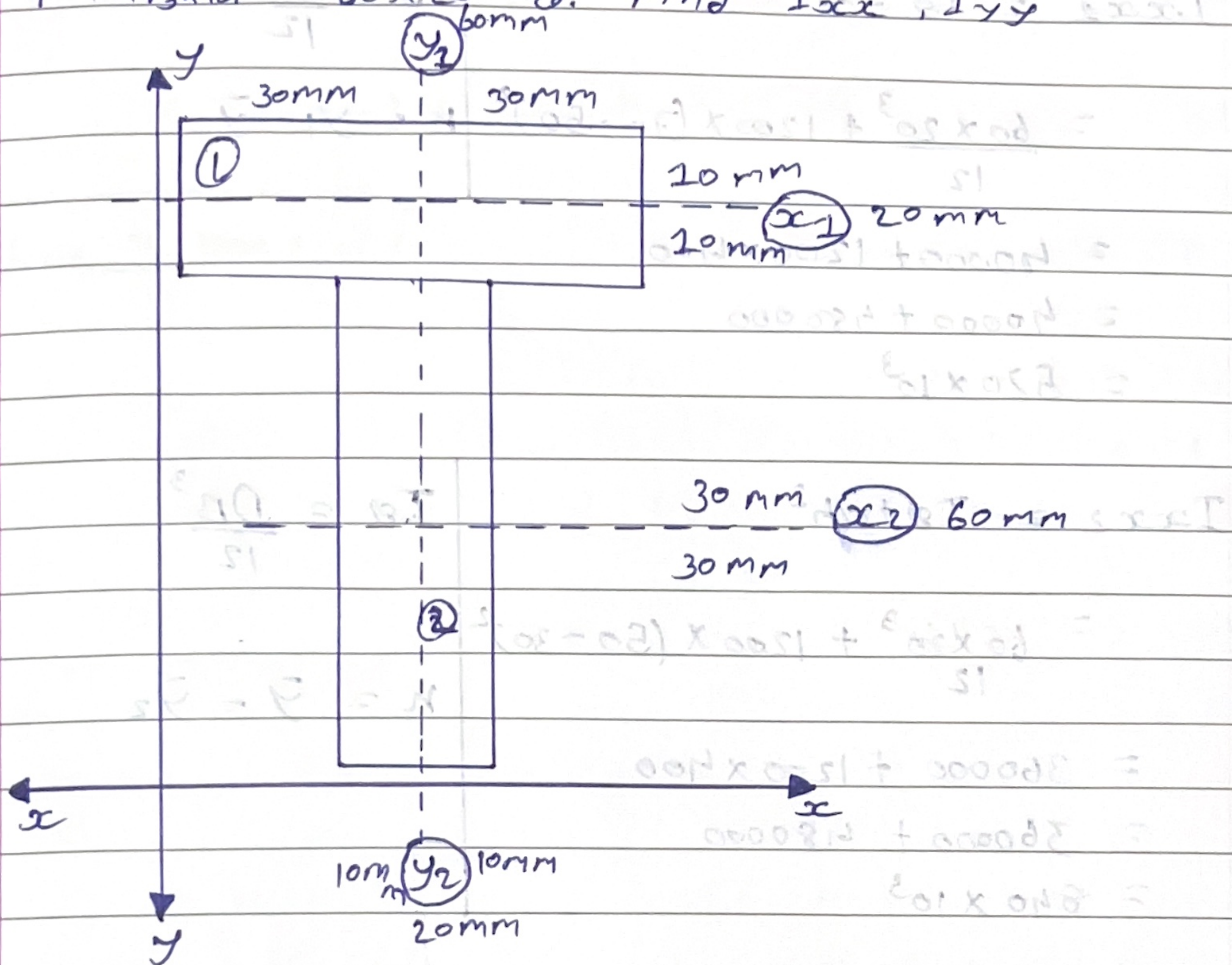
$$\therefore h = \bar{y} - y_2$$

$$I_{xx} = I_{xx1} + I_{xx2}$$

Same for  $I_{yy}$



Ex-2 T-shaped box 20 ed. find  $I_{xx}$ ,  $I_{yy}$



Part :- 1

$$a_1 : 60 \times 20 = 1200 \text{ mm}^2$$

$$x_1 : \frac{60}{2} = 30 \text{ mm} \therefore x_1 = y \text{ to } y_1$$

$$y_2 : 60 + 10 = 70 \text{ mm} \therefore y_2 = x \text{ to } x_2$$

Part :- 2

$$a_2 : 60 \times 20 = 1200 \text{ mm}^2$$

$$y_2 : \frac{60}{2} = 30 \text{ mm} \therefore y_2 = x \text{ to } x_2$$

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2}{a_1 + a_2} = \frac{(1200 \times 70) + (1200 \times 30)}{1200 + 1200} = \frac{84000 + 36000}{2400}$$

$$\bar{y} = 50$$



$$I_{xx1} = I_g + ah^2$$

$$= \frac{60 \times 20^3}{12} + 1200 \times (70 - 50)^2$$

$$= 40000 + 1200 \times 400$$

$$= 40000 + 480000$$

$$= 520 \times 10^3$$

$$I_g = \frac{BD^3}{12}$$

$$h = y_1 - \bar{y}$$

$$I_{xx2} = I_g + ah^2$$

$$= \frac{20 \times 60^3}{12} + 1200 \times (50 - 30)^2$$

$$= 360000 + 1200 \times 400$$

$$= 360000 + 480000$$

$$= 840 \times 10^3$$

$$I_g = \frac{DB^3}{12}$$

$$h = \bar{y} - \bar{y}_2$$

$$I_{xx} = I_{xx1} + I_{xx2}$$

$$= 520 \times 10^3 + 840 \times 10^3$$

$$I_{xx} = 1.36 \times 10^6 \text{ mm}^4$$

$$I_{yy1} = I_g + ah^2$$

$$= \frac{20 \times 60^3}{12} + 1200 \times 0$$

$$= 360 \times 10^3$$

$$I_g = \frac{DB^3}{12}$$

$$h = 0$$

$$I_{yy2} = I_g + ah^2$$

$$= \frac{60 \times 20^3}{12} + 1200 \times 0 = 40 \times 10^3$$

$$I_g = \frac{BD^3}{12}$$

$$h = 0$$



$$I_{yy} = I_{yy1} + I_{yy2}$$

$$= 360 \times 10^3 + 40 \times 10^3$$

$$I_{yy} = 400 \times 10^3$$



ch-3

Imp For SF & BM Reactions

Taking moment @ A,

(RD or RB (last beam support))

$$= RD \times \text{Total Distance of Beam} = (\text{Nearest load from A} \times \text{load to A Distance}) + (\text{load 2 from A} \times \text{load to A Distance})$$

$$= RA = \text{Total vertical load} - RD$$

SF

$$= SFAL = 0 \text{ kN}$$

$$SFAR = SFAL \pm RA$$

$$SFB L = SFAR \pm 0 \text{ kN (load on Left side of A)}$$

$$SFB R = SFB L \pm \text{load on B or Right side of B}$$

$$SFC L = SFB R \pm 0 \text{ (load on Left side of C)}$$

$$SFC R = SFC L \pm \text{load on C or Right side of C}$$

$$SFD L = SFC R \pm 0 \text{ (load on Left side of D)}$$

$$SFD R = SFD L \pm RD = 0$$

BM

First and Last BM = 0

$$BMA = 0$$

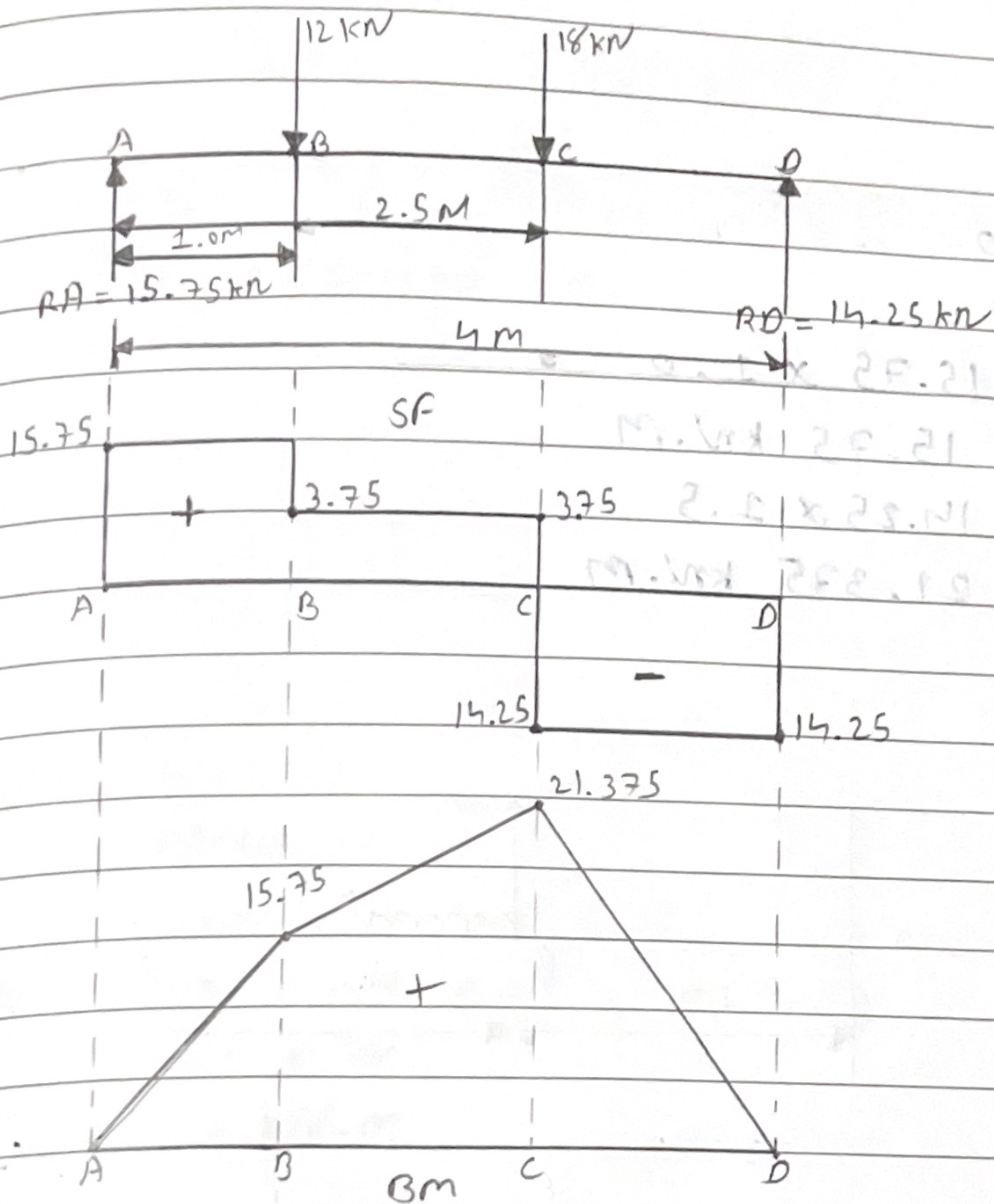
$$BM B = \text{Nearest Reaction} \times \text{Distance Between Reaction}$$

$$BM C = \text{Nearest Reaction} \times \text{Distance Between Reaction}$$

$$BMD = 0$$



Ex-2



- Taking moment @ A

$$= R_D \times 4 = (12 \times 1) + (18 \times 2.5)$$

$$= R_D \times 4 = 57$$

$$= R_D = \frac{57}{4}$$

$$= \boxed{R_D = 14.25}$$

$$\therefore R_A = \text{Total load} - R_D$$

$$= (12 + 18) - 14.25$$

$$\boxed{R_A = 15.75}$$

SF

$$\text{SF at A} = 0 + 15.75 = 15.75 \text{ kN}$$

$$\text{SF at B} = 15.75 - 12 = 3.75 \text{ kN}$$

$$\text{SF at C} = 3.75 - 18 = -14.25 \text{ kN}$$

$$\text{SF at D} = -14.25 + 14.25 = 0 \text{ kN}$$



# BM

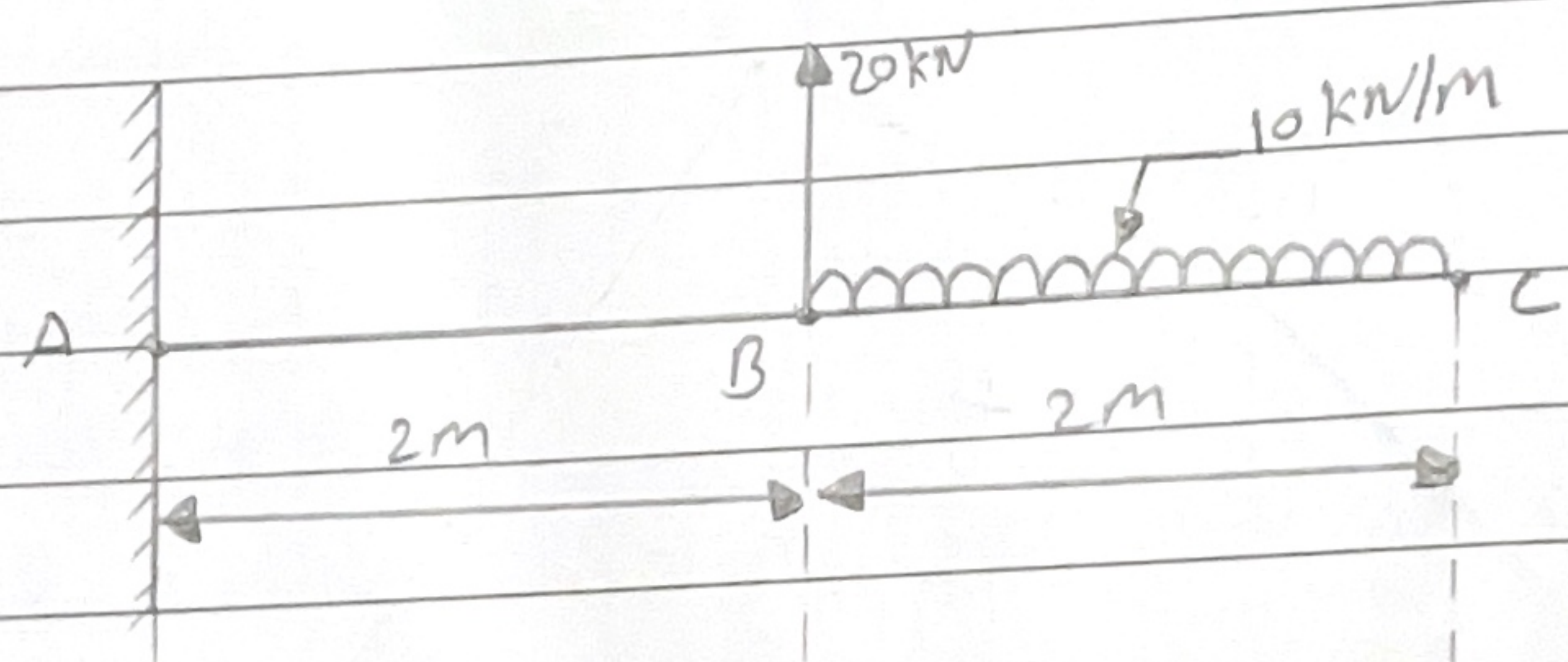
$$BM\ A = 0$$

$$BM\ D = 0$$

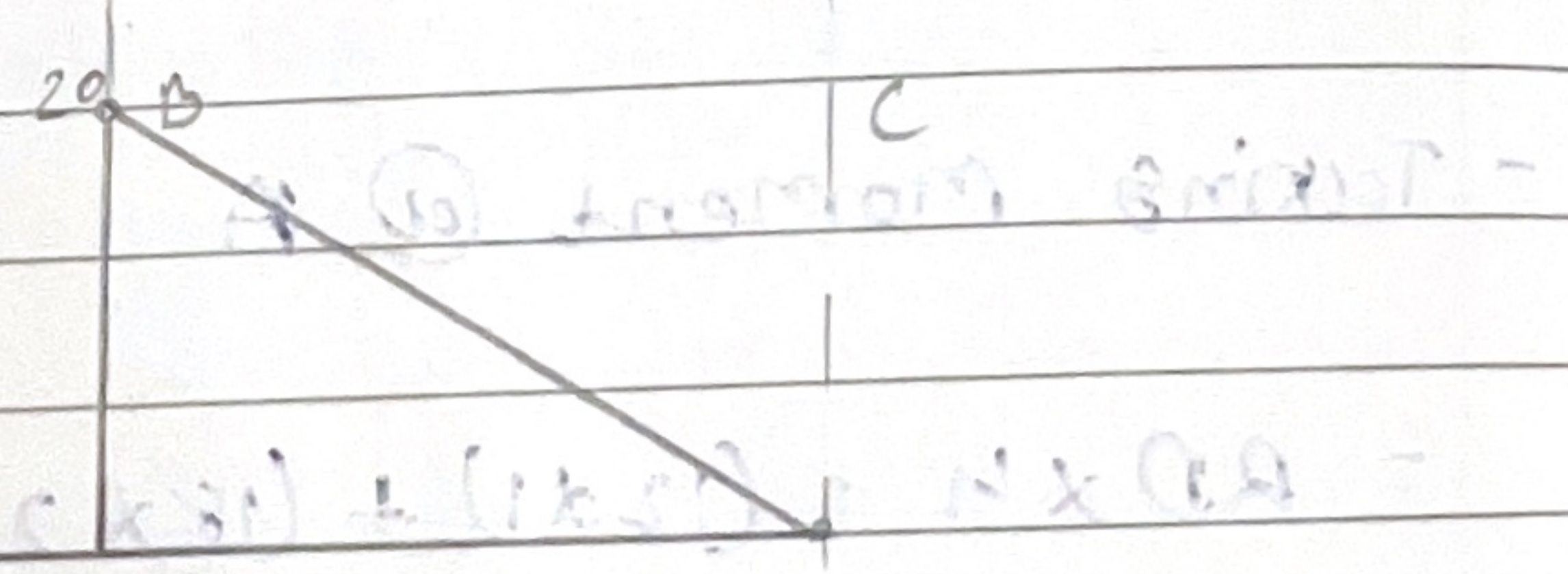
$$BM\ B = 15.75 \times 1.0 = 15.75\ kN.m$$

$$BM\ C = 14.25 \times 1.5 = 21.375\ kN.m$$

## Ex-2

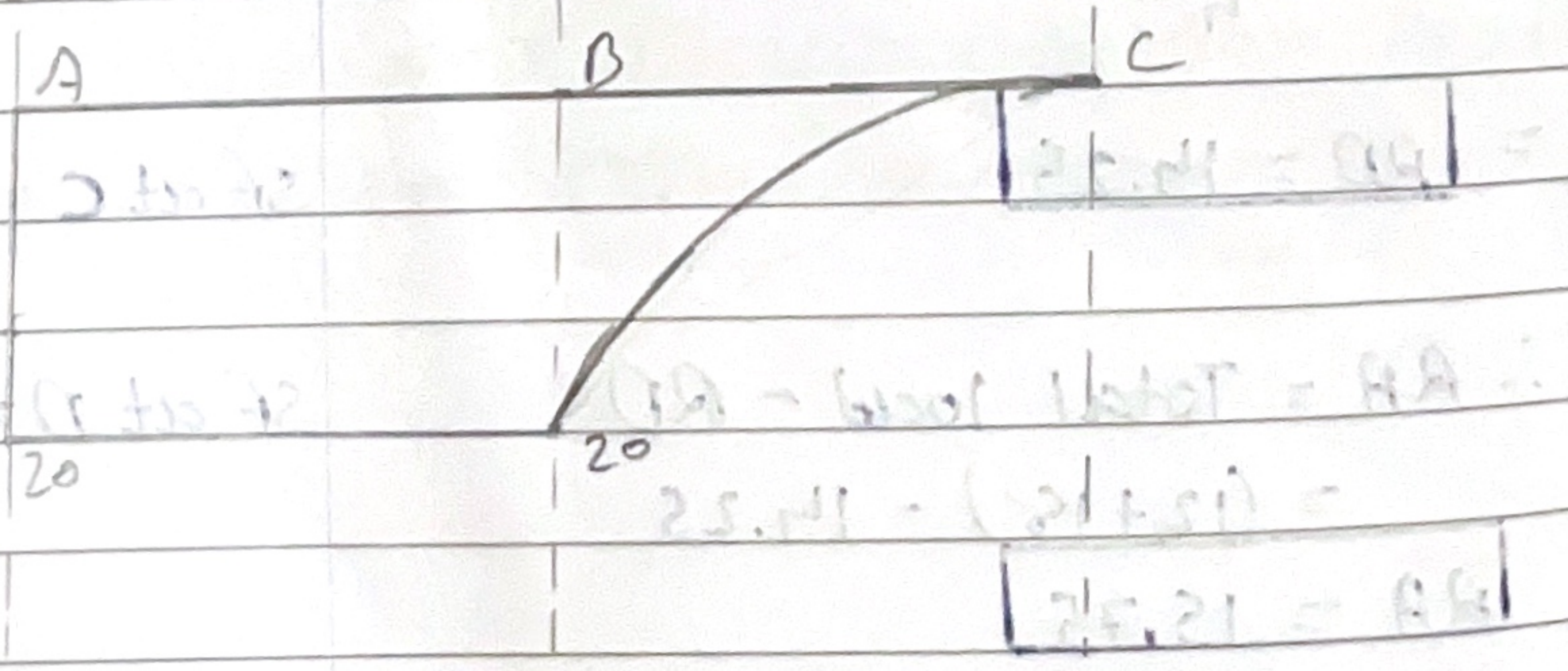


## SF



$$\sum F_v = 0 \Rightarrow -20 + 20 = 0 \quad \text{or} \quad \sum F_h = 0 \Rightarrow 20 - 20 = 0$$

## BM



$$\sum M_A = 0 \Rightarrow -20 \times 2 + 20 \times 2 = 0$$

$$\sum M_B = 0 \Rightarrow -20 \times 1 + 20 \times 1 = 0$$

$$\sum M_C = 0 \Rightarrow -20 \times 2 + 20 \times 2 = 0$$

$$21.375 - (20 \times 1.5) = 0 \Rightarrow 21.375 = 30$$



SF

$$SF \text{ at } C = 0$$

$$SF \text{ at } B = 10 \times 2 - 20 = 0$$

$$SF \text{ at } A = 0$$

BM

$$BM \text{ at } C = 0$$

$$\begin{aligned} BM \text{ at } B &= -10 \times 2 \times 1 \\ &= -20 \text{ kN.m} \end{aligned}$$

$$\begin{aligned} BM \text{ at } A &= -10 \times 2 \times 3 + 20 \times 2 \\ &= -60 + 40 \\ &= -20 \text{ kN.m} \end{aligned}$$



Ch-4Imp For Bending stresses in Beams

$$\frac{M}{I} = \frac{f}{y}$$

$M$  = Moment of resistance or B.M.

$I$  = M.I. about x-x axis

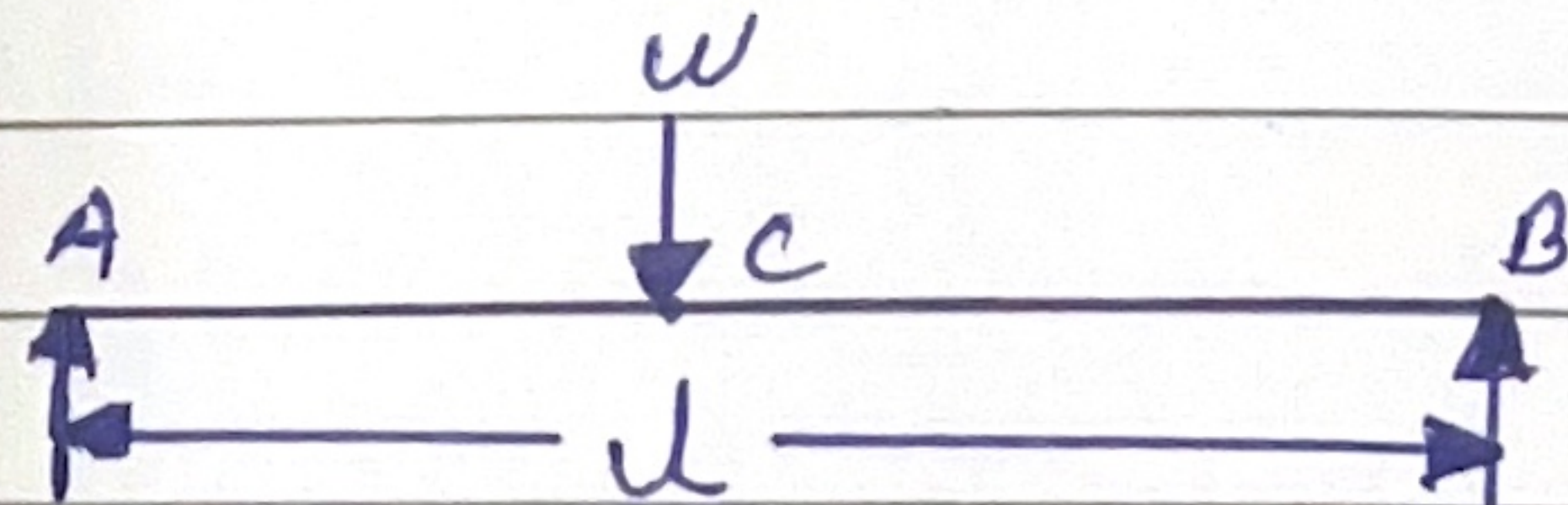
$f$  = Bending stress

$y$  = NA on & 2nd side of axis

• moment of Resistance :

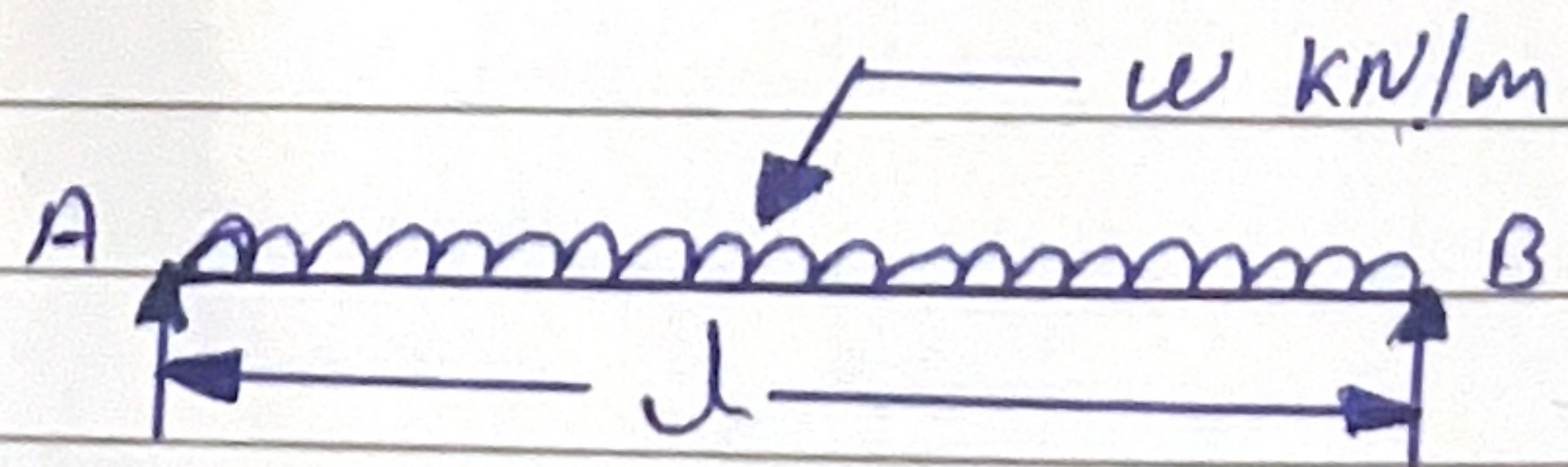
1) simply supported beam with point load:

$$M = \frac{wl}{4}$$



2) simply supported Beam with UDL load:

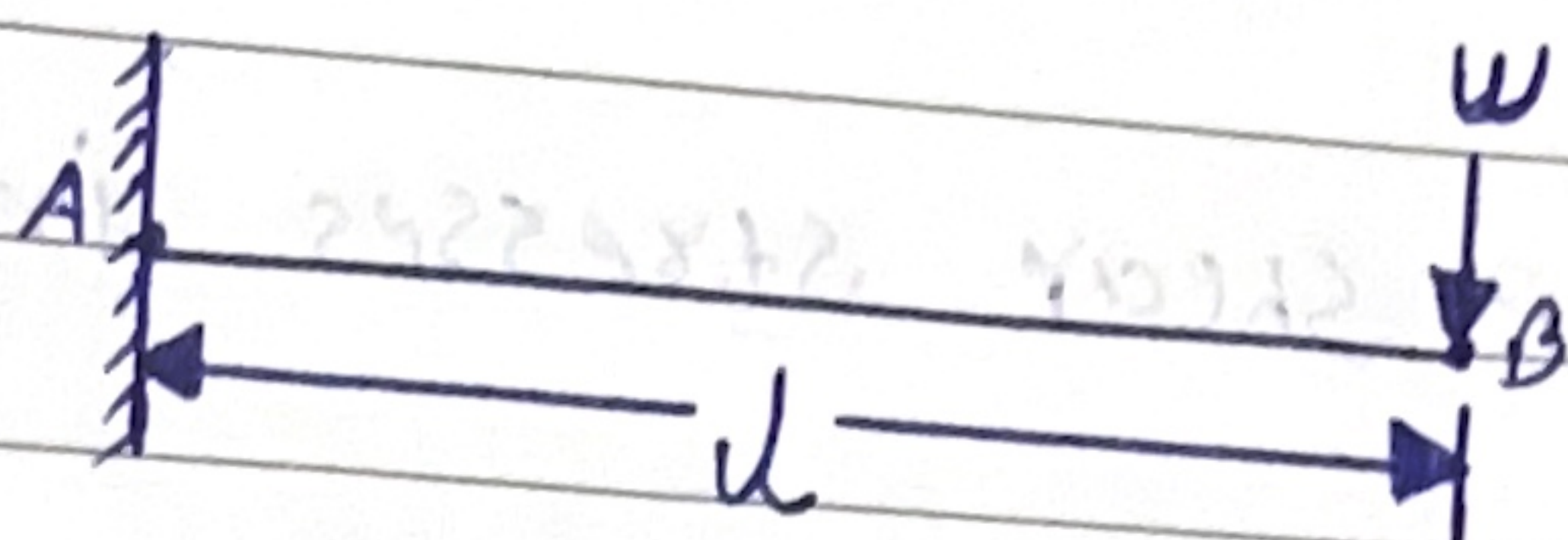
$$M = \frac{wl^2}{8}$$





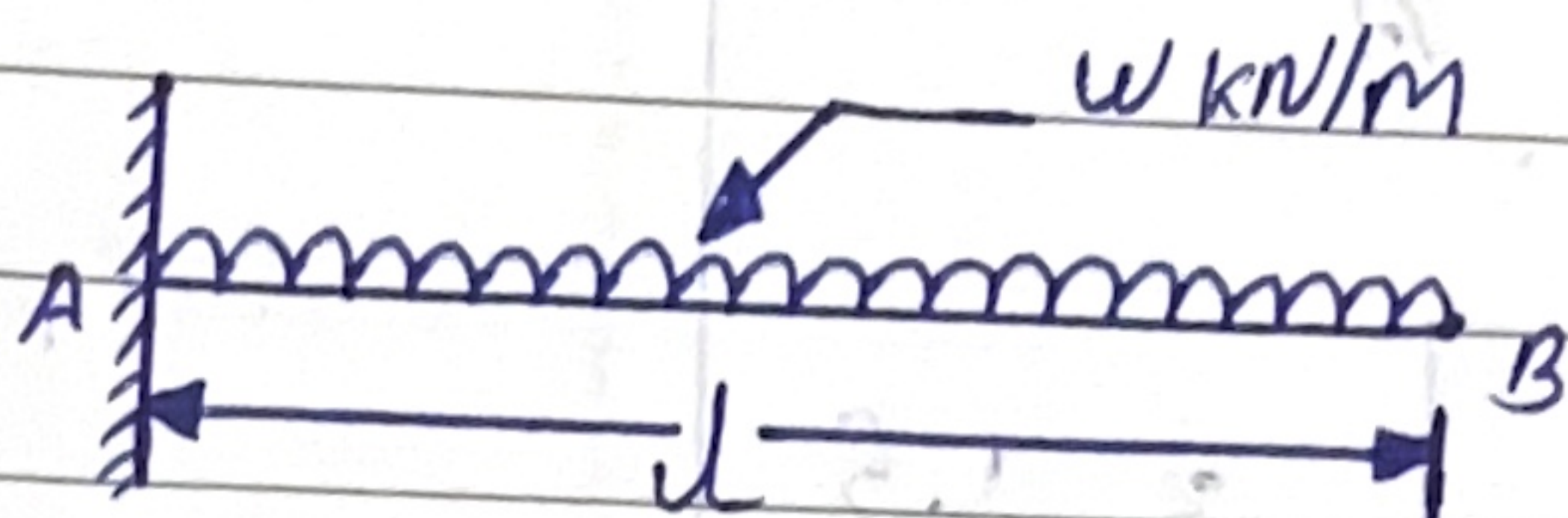
3) cantilever Beam with Point load:

$$M = W \cdot L$$

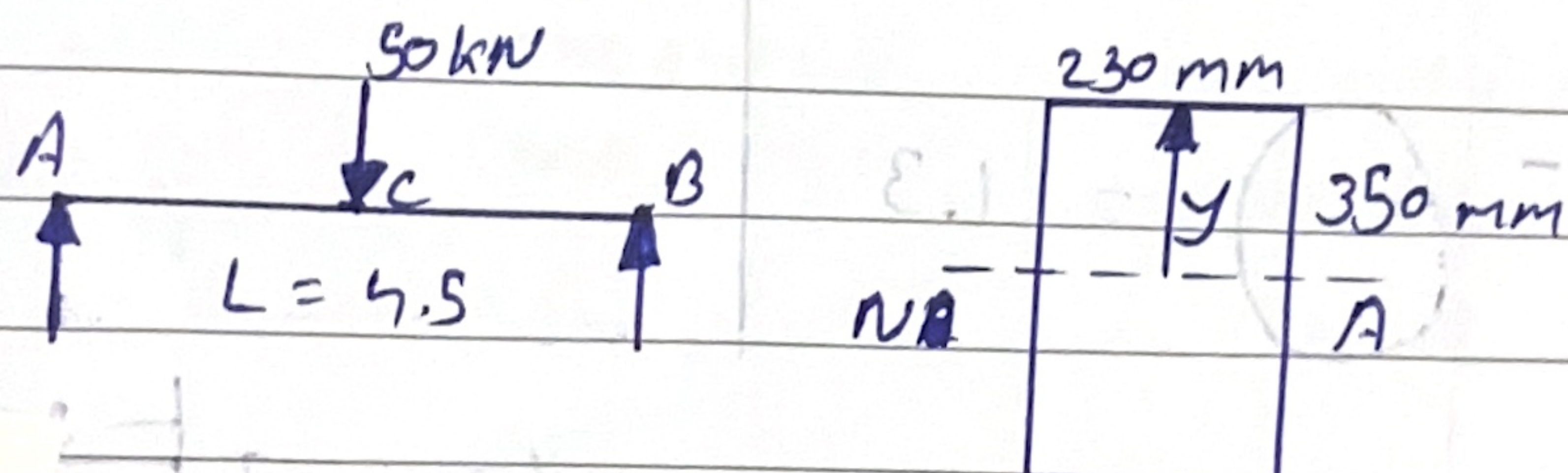


4) cantilever Beam with UDL load:

$$M = \frac{wL^2}{2}$$



Ex-2  $L = 4.5 \text{ m}$   
 $W = 50 \text{ kN}$



$$M = \frac{WL}{4}$$

$$= \frac{50 \times 4.5}{4}$$

$$= 56.25 \text{ kN.m}$$

$$= 56.25 \times 10^6 \text{ N.mm}$$

$$I = \frac{BD^3}{12}$$

$$= \frac{230 \times 350^3}{12}$$

$$= 821.77 \times 10^6 \text{ mm}^4$$

$$y = \frac{350}{2}$$

$$= 175 \text{ mm}$$

$$\frac{M}{I} = \frac{F}{y}$$

$$= \frac{56.25 \times 10^6}{821.77 \times 10^6} = \frac{F}{175}$$

$$= F = \frac{(56.25 \times 10^6) \times (175)}{(821.77 \times 10^6)}$$

$$= \boxed{F = 11.98 \text{ N/mm}^2}$$



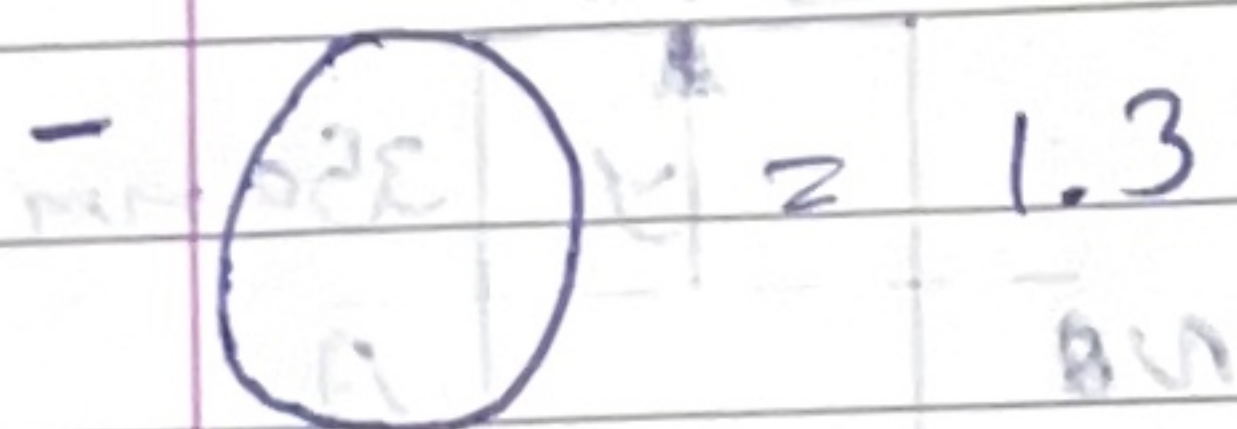
Unit 4-5

Imp shear stresses in Beam:

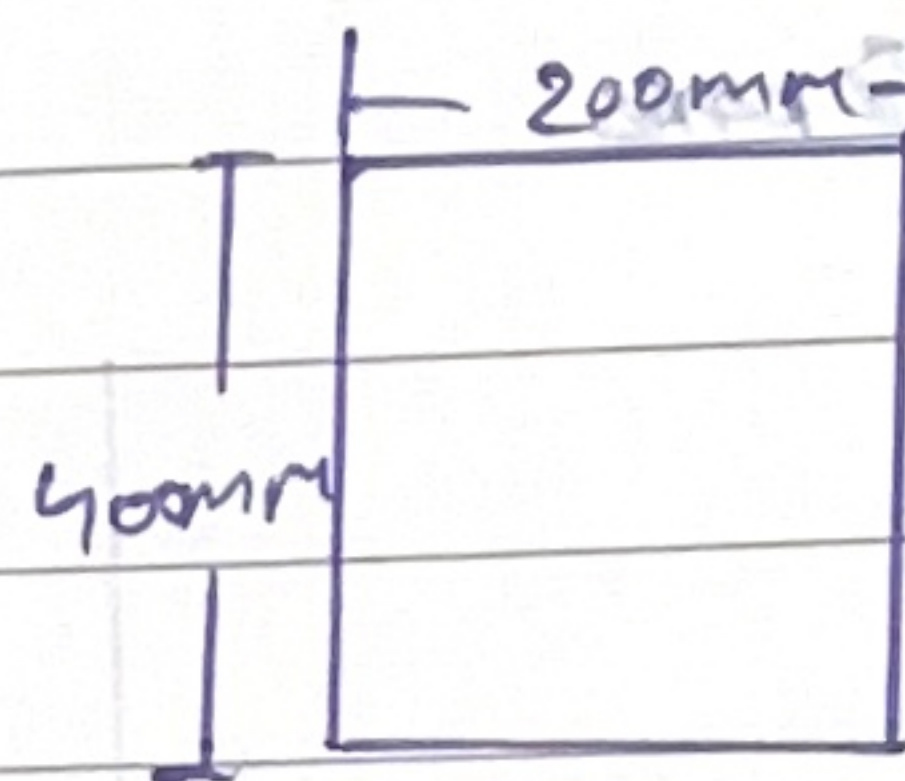
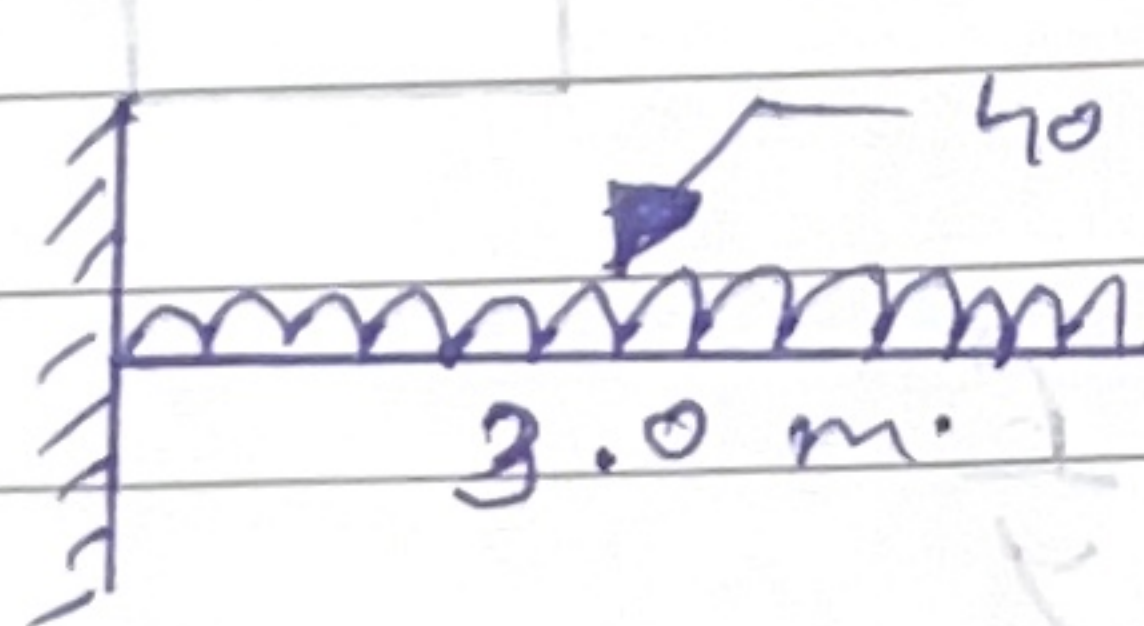
$$\tau_{avg} = 6$$

$$\tau = \frac{P}{A}$$

$$\tau_{max} = 6 \times (1.5 \text{ or } 1.3)$$



Ex-1:



$$F = w \cdot l = 40 \times 3$$

$$= 120 \text{ (kN} \times 25.22)$$

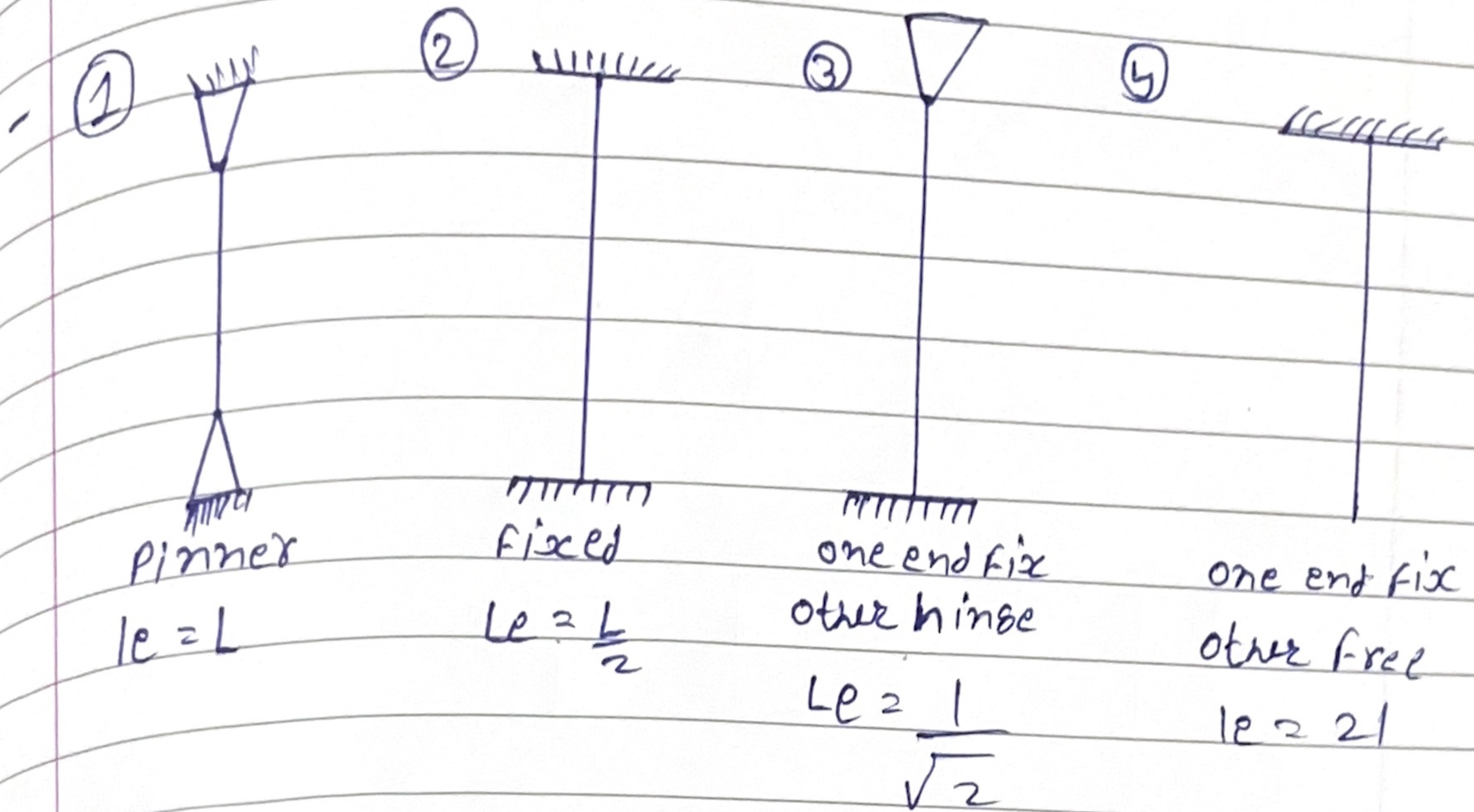
$$A = 200 \times 400$$

$$= 80000 \text{ mm}^2$$

$$\tau_{avg} = \frac{P}{A} = \frac{120 \times 10^3}{80000} = 1.5 \text{ N/mm}^2$$

$$\tau_{max} = 1.5 \times 1.5 = 2.25 \text{ N/mm}^2$$



Ch-8Imp column and struts:

$$P_c = \frac{\pi^2 EI}{(l_e)^2}$$

ex-1)  $d = 400 \text{ mm}$

$$I = \frac{\pi d^4}{64} = 1.256 \times 10^9 \text{ mm}^4$$

7m

$$l = 5 \text{ m}, E = 2 \times 10^5 \text{ N/mm}^2$$

Both ends fixed:

$$\therefore l_e = \frac{l}{2} = \frac{5000}{2}$$

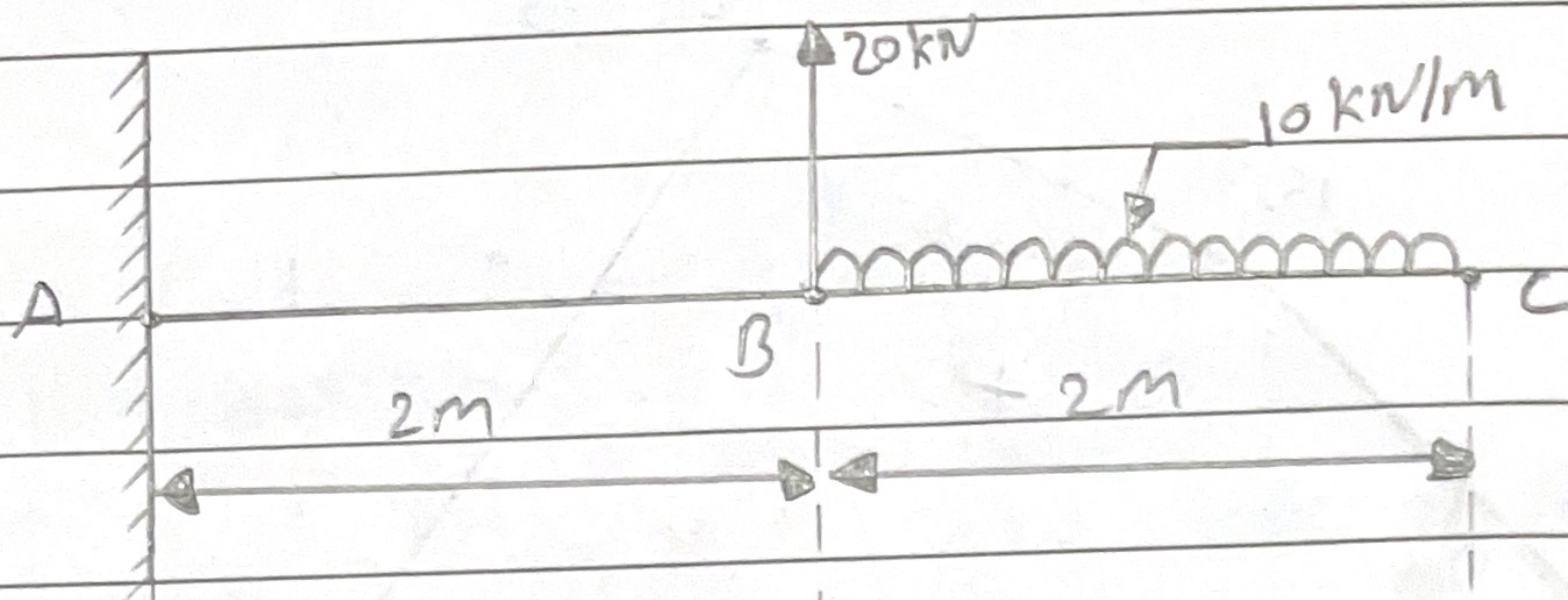
$$l_e = 2500$$

$$P_c = \frac{\pi^2 EI}{(l_e)^2}$$

$$P_c = \frac{\pi \times 2 \times 10^5 \times 1.256 \times 10^9}{(2500)^2}$$

$$= 396.67 \times 10^6 \text{ N}$$





SF

20

B

C

① to find reaction at A

$$\sum F_x = 0 \Rightarrow R_A = 0$$

$$\sum F_y = 0 \Rightarrow R_A = 20 + 20 = 40 \text{ kN}$$

$$\sum M_A = 0$$

BM

A

B

C

$$\sum M_A = 0$$

$$20 \times 2 + 20 \times 2 = R_A \times 4$$

$$40 + 40 = R_A$$

20

$$R_A = 80 \text{ kN}$$

$$20 \times 2 + 20 \times 2 = R_A \times 4$$

$$R_A = 80 \text{ kN}$$